

SUMMARY OF PRODUCT CHARACTERISTICS

1 NAME OF THE MEDICINAL PRODUCT

ZAHRON ASA (Rosuvastatin 5mg/Acetylsalicylic acid 100mg) Hard Capsule
ZAHRON ASA (Rosuvastatin 10mg/Acetylsalicylic acid 100mg) Hard Capsule
ZAHRON ASA (Rosuvastatin 20mg/Acetylsalicylic acid 100mg) Hard Capsule

2 QUALITATIVE AND QUANTITATIVE COMPOSITION

5 mg/100 mg: each hard capsule contains 5 mg rosuvastatin (as rosuvastatin calcium) and 100 mg acetylsalicylic acid.

10 mg/100 mg: each hard capsule contains 10 mg rosuvastatin (as rosuvastatin calcium) and 100 mg acetylsalicylic acid.

20 mg/100 mg: each hard capsule contains 20 mg rosuvastatin (as rosuvastatin calcium) and 100 mg acetylsalicylic acid.

Excipient with known effect:

Each 5 mg/100 mg, capsule, hard contains 25.92 mg lactose monohydrate, see section 4.4.

Each 10 mg/100 mg, capsule, hard contains 51.84 mg lactose monohydrate, see section 4.4.

Each 20 mg/100 mg, capsule, hard contains 103.68 mg lactose as monohydrate, see section 4.4.

For the full list of excipients, see section 6.1.

3 PHARMACEUTICAL FORM

Capsule, hard.

5 mg/100 mg: hard gelatin capsules of size 2 and white opaque body and dark green opaque cap. Each capsule contains one white or almost white, biconvex, oval uncoated tablet of acetylsalicylic acid and one film-coated, brown, biconvex, round tablet of rosuvastatin 5 mg.

10 mg/100 mg: hard gelatin capsules of size 1 and white opaque body with black overprint "ASA 100" and light green opaque cap with black overprint "RSV 10". Each capsule contains one white or almost white, biconvex, oval uncoated tablet of acetylsalicylic acid and one film-coated, brown, biconvex, round tablet of rosuvastatin 10 mg.

20 mg/100 mg: hard gelatin capsules of size 0 and white opaque body with black overprint "ASA 100" and green opaque cap with black overprint "RSV 20". Each capsule contains one white or almost white, biconvex, oval uncoated tablet of acetylsalicylic acid and two film-coated, brown, biconvex, round tablets of rosuvastatin 10 mg.

4 CLINICAL PARTICULARS

4.1 Therapeutic Indications

ZAHRON ASA is indicated for the secondary prevention of cardiovascular events as substitution therapy in adult patients adequately controlled with the monocomponents given concomitantly at equivalent therapeutical doses.

4.2 Posology and method of administration

Posology

Adults

The recommended dose of ZAHRON ASA is one capsule per day.

The fixed dose combination is not suitable for initial therapy.

Before switching to ZAHRON ASA patients should be controlled on stable doses of the monocomponents taken at the same time. The dose of ZAHRON ASA should be based on the doses of the individual components of the combination at the time of switching.

If the change of posology is required for any of the active substances of the fixed combination due to any reason (e.g. newly diagnosed related illness, change of the condition of the patient or due to drug interaction), the individual components should be used again in order to determine the posology.

Older patients

No dosage adjustment is required.

Patients with renal impairment

No dosage adjustment is required for patients with mild to moderate renal impairment.

The use of ZAHRON ASA in patients with severe renal impairment is contraindicated for all doses. (See section 4.3 and section 5.2).

Patients with hepatic impairment

No dosage adjustment is required for patients with mild to moderate hepatic impairment.

The use of ZAHRON ASA in patients with severe hepatic impairment is contraindicated for all doses.

There was no increase in systemic exposure to rosuvastatin in subjects with Child-Pugh scores of 7 or below. However, increased systemic exposure has been observed in subjects with Child-Pugh scores of 8 and 9 (see section 5.2). In these patients an assessment of renal function should be considered (see section 4.4). There is no experience in subjects with Child-Pugh scores above 9. ZAHRON ASA is contraindicated in patients with active liver disease (see section 4.3).

Ethnic Differences

Increased systemic exposure to rosuvastatin has been seen in Asian subjects (see section 5.2).

Genetic polymorphisms

Specific types of genetic polymorphisms are known that can lead to increased rosuvastatin exposure (see section 5.2). For patients who are known to have such specific types of polymorphisms, a lower daily dose of rosuvastatin is recommended.

Dosage in patients with pre-disposing factors to myopathy

The recommended start dose of rosuvastatin is 5 mg in patients with pre-disposing factors to myopathy.

Concomitant therapy

Rosuvastatin is a substrate of various transporter proteins (e.g. OATP1B1 and BCRP). The risk of myopathy (including rhabdomyolysis) is increased when rosuvastatin is administered concomitantly with certain medicinal products that may increase the plasma concentration of rosuvastatin due to interactions with these transporter proteins (e.g. ciclosporin and certain protease inhibitors including combinations of ritonavir with atazanavir, lopinavir, and/or tipranavir; see sections 4.4 and 4.5).

Whenever possible, alternative medications should be considered, and, if necessary, consider temporarily discontinuing rosuvastatin therapy. In situations where co-administration of these medicinal products with rosuvastatin is unavoidable, the benefit and the risk of concurrent treatment

and dosing adjustments of rosuvastatin should be carefully considered (see section 4.5). Higher doses of acetylsalicylic acid diminish the ameliorating effects of rosuvastatin, therefore taking additional doses of acetylsalicylic acid should be avoided (see section 4.5).

Paediatric population

The safety and efficacy of rosuvastatin/acetylsalicylic acid in children and adolescents below 18 years have not been established. Rosuvastatin/acetylsalicylic acid is not recommended for use in patients below 18 years.

Method of administration

ZAHRON ASA can be taken at any time of the day and should be taken with food. Capsules should be swallowed with plenty of liquid and should not be chewed.

4.3 Contraindications

Linked to rosuvastatin component:

- Active liver disease including unexplained, persistent elevations of serum transaminases and any serum transaminase elevation exceeding 3 x the upper limit of normal (ULN).
- Severe renal impairment (creatinine clearance <30 ml/min).
- Myopathy.
- In patients receiving concomitant combination of sofosbuvir/velpatasvir/voxilaprevir (see section 4.5).
- Concomitant ciclosporin treatment.
- Pregnancy and lactation and in women of childbearing potential not using appropriate contraceptive measures.
- Hypersensitivity to rosuvastatin.

Linked to acetylsalicylic acid component:

- Hypersensitivity to acetylsalicylic acid. Hypersensitivity to non-steroidal anti-inflammatory drugs (NSAIDs); patients with asthma, rhinitis, and nasal polyps; patients with pre-existing mastocytosis, in whom the use of acetylsalicylic acid may induce severe hypersensitivity reactions (including circulatory shock with flushing, hypotension, tachycardia and vomiting).
- Active, or history of recurrent peptic ulcer and/or gastric/intestinal haemorrhage, or other kinds of bleeding such as cerebrovascular haemorrhages.
- Haemorrhagic diathesis, coagulation disorders such as haemophilia and thrombocytopenia.
- Severe hepatic or renal impairment.
- Severe uncompensated heart failure.
- Methotrexate used at doses >15 mg/week (see section 4.5).
- Doses > 100 mg acetylsalicylic acid/day during the third trimester of pregnancy (see section 4.6).

Linked to ZAHRON ASA:

- Hypersensitivity to any of the excipients listed in section 6.1.

ZAHRON ASA contains soya oil.

If you are allergic to peanut or soya, do not use this medicinal product.

4.4 Special warnings and precautions for use

Rosuvastatin

Renal effects

Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with higher doses of rosuvastatin, in particular 40 mg, where it was transient or intermittent in most cases. Proteinuria has not been shown to be predictive of acute or progressive renal disease (see section 4.8). The reporting rate for serious renal events is higher at the 40 mg dose.

Skeletal muscle effects

Effects on skeletal muscle e.g. myalgia, myopathy and, rarely, rhabdomyolysis have been reported in rosuvastatin- treated patients with all doses and in particular with doses >20 mg. Very rare cases of rhabdomyolysis have been reported with the use of ezetimibe in combination with HMG-CoA reductase inhibitors. A pharmacodynamic interaction cannot be excluded (see section 4.5) and caution should be exercised with their combined use.

Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of rosuvastatin and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of rosuvastatin with fibrates or niacin should be carefully weighed against the potential risks of such combinations.

Creatine kinase measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of a plausible alternative cause of CK increase which may confound interpretation of the result. If CK levels are significantly elevated at baseline (>5xULN) a confirmatory test should be carried out within 5 – 7 days. If the repeat test confirms a baseline CK >5xULN, treatment should not be started.

Before treatment

ZAHRON ASA, as with other HMG-CoA reductase inhibitor containing products, should be prescribed with caution in patients with pre-disposing factors for myopathy/rhabdomyolysis due to rosuvastatin component. Such factors include:

- Renal impairment
- Hypothyroidism
- Personal or family history of hereditary muscular disorders
- History of muscular toxicity with another HMG-CoA reductase inhibitor or fibrate
- Alcohol abuse
- Age >70 years
- Situations where an increase in plasma levels may occur (see sections 4.2, 4.5 and 5.2)
- Concomitant use of fibrates.

In such patients the risk/benefit ratio of the treatment should be considered and clinical monitoring is recommended. If CK levels are significantly elevated at baseline (>5xULN) treatment should not be started.

Whilst on treatment

Patients should be asked to report inexplicable muscle pain, weakness or cramps immediately, particularly if associated with malaise or fever. CK levels should be measured in these patients. Therapy should be discontinued if CK levels are markedly elevated (>5xULN) or if muscular symptoms are severe and cause daily discomfort (even if CK levels are <5x ULN). If symptoms resolve and CK levels return to normal, then re-introducing rosuvastatin or an alternative HMG-CoA reductase inhibitor at the lowest dose with close monitoring can be reconsidered. Routine monitoring of CK levels in asymptomatic patients is not warranted. There have been very rare reports of an immune- mediated necrotizing myopathy (IMNM) during or after treatment with statins, including rosuvastatin. IMNM is clinically characterized by proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment.

In clinical trials on small number of patients dosed with rosuvastatin and concomitant therapy there was no evidence of increased skeletal muscle effects. However, an increase in the incidence of myositis and myopathy has been seen in patients receiving other HMG-CoA reductase inhibitors together with fibric acid derivatives including gemfibrozil, ciclosporin, nicotinic acid, azole antifungals, protease inhibitors and macrolide antibiotics. Gemfibrozil increases the risk of myopathy when given concomitantly with some HMG-CoA reductase inhibitors. Therefore, the combination of

rosuvastatin/acetylsalicylic acid and gemfibrozil is not recommended. The benefit of further alterations in lipid levels by the combined use of ZAHRON ASA with fibrates or niacin should be carefully weighed against the potential risks of such combinations. (See section 4.5 and section 4.8.)

ZAHRON ASA must not be co-administered with systemic formulations of fusidic acid or within 7 days of stopping fusidic acid treatment. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving fusidic acid and statins in combination (see section 4.5). Patients should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness. Statin therapy may be re-introduced seven days after the last dose of fusidic acid.

In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g., for the treatment of severe infections, the need for co-administration of ZAHRON ASA and fusidic acid should only be considered on a case by case basis and under close medical supervision.

ZAHRON ASA should not be used in any patient with an acute, serious condition suggestive of myopathy or predisposing to the development of renal failure secondary to rhabdomyolysis (e.g. sepsis, hypotension, major surgery, trauma, severe metabolic, endocrine and electrolyte disorders; or uncontrolled seizures).

Daptomycin

Cases of myopathy and/or rhabdomyolysis have been reported with HMG-CoA reductase inhibitors co-administered with daptomycin. Caution should be used when prescribing HMG-CoA reductase inhibitors with daptomycin, as either agent can cause myopathy and/or rhabdomyolysis when given alone. Consideration should be given to temporarily suspend ZAHRON ASA in patients taking daptomycin unless the benefits of concomitant administration outweigh the risk. If co-administration cannot be avoided, CPK levels should be measured more frequently than once weekly and patients should be closely monitored for any signs or symptoms that might represent myopathy.

Severe cutaneous adverse reactions

Severe cutaneous adverse reactions including Stevens-Johnson syndrome (SJS) and drug reaction with eosinophilia and systemic symptoms (DRESS), which could be life-threatening or fatal, have been reported with rosuvastatin (see section 4.8). At the time of prescription, patients should be advised of the signs and symptoms of severe skin reactions and be closely monitored. If signs and symptoms suggestive of this reaction appears, ZAHRON ASA should be discontinued immediately and an alternative treatment should be considered.

If the patient has developed a serious reaction such as SJS or DRESS with the use of ZAHRON ASA, treatment with ZAHRON ASA must not be restarted in this patient at any time.

Liver effects

As with other HMG-CoA reductase inhibitor containing products, ZAHRON ASA should be used with caution in patients who consume excessive quantities of alcohol and/or have a history of liver disease.

It is recommended that liver function tests be carried out prior to, and 3 months following, the initiation of rosuvastatin treatment. ZAHRON ASA should be discontinued or the dose of rosuvastatin reduced if the level of serum transaminases is greater than 3 times ULN.

In patients with secondary hypercholesterolaemia caused by hypothyroidism or nephrotic syndrome, the underlying disease should be treated prior to initiating therapy with ZAHRON ASA.

Ethnic differences

Pharmacokinetic studies show an increase in exposure to rosuvastatin in Asian subjects compared with Caucasians (see section 4.2, section 4.3 and section 5.2).

Protease inhibitors

Increased systemic exposure to rosuvastatin has been observed in subjects receiving rosuvastatin concomitantly with various protease inhibitors in combination with ritonavir. Consideration should be given both to the benefit of lipid lowering by use of rosuvastatin in HIV patients receiving protease inhibitors and the potential for increased rosuvastatin plasma concentrations when initiating and up titrating rosuvastatin doses in patients treated with protease inhibitors. The concomitant use with protease inhibitors is not recommended unless the dose of rosuvastatin is adjusted. (see sections 4.2 and 4.5).

Interstitial lung disease

Exceptional cases of interstitial lung disease have been reported with some statins, especially with long term therapy (see section 4.8). Presenting features can include dyspnea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected that a patient has developed interstitial lung disease, statin therapy should be discontinued.

Diabetes mellitus

Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI > 30 kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

In the JUPITER study, the reported overall frequency of diabetes mellitus was 2.8% in rosuvastatin and 2.3% in placebo, mostly in patients with fasting glucose 5.6 to 6.9 mmol/L.

Paediatric Population

The evaluation of linear growth (height), weight, BMI (body mass index), and secondary characteristics of sexual maturation by Tanner staging in paediatric patients 6 to 17 years of age taking rosuvastatin is limited to a two-year period. After two years of study treatment, no effect on growth, weight, BMI or sexual maturation was detected (see section 5.1).

In a clinical trial of children and adolescents receiving rosuvastatin for 52 weeks, CK elevations >10xULN and muscle symptoms following exercise or increased physical activity were observed more frequently compared to observations in clinical trials in adults (see section 4.8).

Myasthenia gravis or ocular myasthenia

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. ZAHRON ASA should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

Acetylsalicylic acid (ASA)

Concomitant treatment with anticoagulants (coumarin derivatives, heparin) is not recommended and should generally be avoided. If concurrent use cannot be avoided, frequent monitoring of the International Normalization Ratio (INR) is indicated and patients should be cautioned to watch for signs of bleeding, especially in the gastrointestinal tract.

Caution should be taken in patients with hypersensitivity to other analgesics / anti-inflammatories / anti-rheumatics or other allergenic substances (see section 4.3).

Close medical monitoring is also necessary for patients with pre-existing allergies (e.g. skin, pruritus, urticaria), asthma, hay fever, swollen nasal mucosa (nasal polyps) or chronic respiratory disease.

Patients with a history of gastrointestinal ulcer disease and/or gastrointestinal hemorrhages should avoid using ASA (which can cause gastric mucosal irritation and bleeding).

Caution should be exercised in patients with hepatic insufficiency (as ASA is metabolised mainly via the liver, see section 5.2).

In patients with impaired renal function or with impaired cardiovascular circulation (e.g. renal vascular disease, congestive heart failure, volume depletion, major surgery, sepsis or major hemorrhagic events): acetylsalicylic acid may further increase the risk of renal impairment or acute renal failure.

At low doses, acetylsalicylic acid reduces the excretion of uric acid. This may trigger gout attacks in susceptible patients.

The concomitant administration of this active substance with uricosuric agents like benzbromarone, probenecid, sulphinpyrazone is not recommended (see section 4.5).

ASA must be used with care in cases of very severe menstrual bleeding.

It is preferable to stop use of ASA before a surgical procedure (including tooth extraction) because of the risk of a prolonged bleeding time or an aggravation of the bleeding. The length of the interruption of the treatment should be determined on a case-by-case basis, but will usually be one week.

There is possible association between ASA and Reye's syndrome when given to children. Reye's syndrome is a very rare disease, which affects the brain and liver, and can be fatal. ZAHRON ASA should not be given to children or adolescents with febrile illnesses except where specifically indicated by a physician and only if other measures are ineffective. (see section 4.2).

In patients with severe glucose-6-phosphate dehydrogenase deficiency: acetylsalicylic acid can induce haemolysis or haemolytic anaemia. Factors that may increase the risk of haemolysis include high dosage, fever or acute infections.

Alcohol may increase the risk of gastrointestinal injury when taken with ASA. Patients should be counseled about the risks of gastrointestinal injury and bleeding while taking Rosuvastatin plus ASA with alcohol, especially if alcohol consumption is chronic or heavy.

Lactose

Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

4.5 Interaction with other medicinal products and other forms of interaction

Linked to rosuvastatin component

Effect of co-administered medicinal products on rosuvastatin

Transporter protein inhibitors: Rosuvastatin is a substrate for certain transporter proteins including the hepatic uptake transporter OATP1B1 and efflux transporter BCRP. Concomitant administration of rosuvastatin with medicinal products that are inhibitors of these transporter proteins may result in increased rosuvastatin plasma concentrations and an increased risk of myopathy (see sections 4.2, 4.4, and 4.5 Table 1).

Ciclosporin: During concomitant treatment with rosuvastatin and ciclosporin, rosuvastatin AUC values were on average 7 times higher than those observed in healthy volunteers (see Table 1). Rosuvastatin is contraindicated in patients receiving concomitant ciclosporin (see section 4.3). Concomitant administration did not affect plasma concentrations of ciclosporin.

Protease inhibitors: Although the exact mechanism of interaction is unknown, concomitant protease inhibitor use may strongly increase rosuvastatin exposure (see Table 1). For instance, in a pharmacokinetic study, co-administration of 10 mg rosuvastatin and a combination product of two protease inhibitors (300 mg atazanavir / 100 mg ritonavir) in healthy volunteers was associated with an approximately three-fold and seven-fold increase in rosuvastatin steady-state AUC and $C_{\max}$ respectively. The concomitant use of rosuvastatin and some protease inhibitor combinations may be considered after careful consideration of rosuvastatin dose adjustments based on the expected increase in rosuvastatin exposure (see sections 4.2, 4.4, and 4.5 Table 1).

Gemfibrozil and other lipid-lowering products: Concomitant use of rosuvastatin and gemfibrozil resulted in a 2-fold increase in rosuvastatin $C_{\max}$ and AUC (see section 4.4). Based on data from specific interaction studies no pharmacokinetic relevant interaction with fenofibrate is expected, however a pharmacodynamic interaction may occur. Gemfibrozil, fenofibrate, other fibrates and lipid lowering doses ($>$ or equal to 1g/day) of niacin (nicotinic acid) increase the risk of myopathy when given concomitantly with HMG- CoA reductase inhibitors, probably because they can produce myopathy when given alone.

Ezetimibe: Concomitant use of 10 mg rosuvastatin and 10 mg ezetimibe resulted in a 1.2-fold increase in AUC of rosuvastatin in hypercholesterolaemic subjects (Table 1). However, a pharmacodynamic interaction, in terms of adverse effects, between ZAHRON ASA and ezetimibe cannot be ruled out (see section 4.4).

Daptomycin: The risk of myopathy and/or rhabdomyolysis may be increased by concomitant administration of HMG-CoA reductase inhibitors and daptomycin. Consideration should be given to suspending ZAHRON ASA temporarily in patients taking daptomycin unless the benefits of concomitant administration outweigh the risk. If co-administration cannot be avoided, CPK levels should be measured more frequently than once weekly and patients should be closely monitored for any signs or symptoms that might represent myopathy (see section 4.4).

Antacid: The simultaneous dosing of rosuvastatin with an antacid suspension containing aluminium and magnesium hydroxide resulted in a decrease in rosuvastatin plasma concentration of approximately 50%. This effect was mitigated when the antacid was dosed 2 hours after ZAHRON ASA. The clinical relevance of this interaction has not been studied.

Erythromycin: Concomitant use of rosuvastatin and erythromycin resulted in a 20% decrease in AUC and a 30% decrease in $C_{\max}$ of rosuvastatin. This interaction may be caused by the increase in gut motility caused by erythromycin.

Cytochrome P450 enzymes: Results from *in vitro* and *in vivo* studies show that rosuvastatin is neither an inhibitor nor an inducer of cytochrome P450 isoenzymes. In addition, rosuvastatin is a poor substrate for these isoenzymes. Therefore, drug interactions resulting from cytochrome P450-mediated metabolism are not expected. No clinically relevant interactions have been observed between rosuvastatin and either fluconazole (an inhibitor of CYP2C9 and CYP3A4) or ketoconazole (an inhibitor of CYP2A6 and CYP3A4).

Ticagrelor: Ticagrelor might affect renal excretion of rosuvastatin, increasing the risk for rosuvastatin accumulation. Although the exact mechanism is not known, in some cases, concomitant use of ticagrelor and rosuvastatin led to renal function decrease, increased CPK level and rhabdomyolysis.

Interactions requiring rosuvastatin dose adjustments (see also Table 1): When it is necessary to co-administer rosuvastatin with other medicinal products known to increase exposure to rosuvastatin, doses of rosuvastatin should be adjusted. Start with a 5 mg once daily dose of rosuvastatin if the expected increase in exposure (AUC) is approximately 2-fold or higher. The maximum daily dose of rosuvastatin should be adjusted so that the expected rosuvastatin exposure would not likely exceed that of a 40 mg daily dose of rosuvastatin taken without interacting medicinal products, for example a

20 mg dose of rosuvastatin with gemfibrozil (1.9-fold increase), and a 10 mg dose of rosuvastatin with combination atazanavir/ritonavir (3.1-fold increase).

If medicinal product is observed to increase rosuvastatin AUC less than 2-fold, the starting dose need not be decreased but caution should be taken if increasing the rosuvastatin dose above 20mg.

Table 1. Effect of co-administered medicinal products on rosuvastatin exposure (AUC; in order of decreasing magnitude) from published clinical trials		
2-fold or greater than 2-fold increase in AUC of rosuvastatin		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Sofosbuvir/velpatasvir/voxilaprevir (400 mg-100 mg-100 mg) + Voxilaprevir (100 mg) once daily for 15 days	10 mg single dose	7.4-fold ↑
Ciclosporin 75 mg BID to 200 mg BID, 6 months	10 mg OD, 10 days	7.1-fold ↑
Darolutamide 600 mg BID, 5 days	5 mg, single dose	5.2-fold ↑
Regorafenib 160 mg, OD, 14 days	5 mg, single dose	3.8-fold ↑
Atazanavir 300 mg/ritonavir 100 mg OD, 8 days	10 mg, single dose	3.1-fold ↑
Roxadustat 200 mg QOD	10 mg, single dose	2.9-fold ↑
Velpatasvir 100 mg OD	10 mg, single dose	2.7-fold ↑
Ombitasvir 25 mg/paritaprevir 150 mg/ Ritonavir 100 mg OD/ dasabuvir 400 mg BID, 14 days	5 mg, single dose	2.6-fold ↑
Teriflunomide	Not available	2.5-fold ↑
Grazoprevir 200 mg/elbasvir 50 mg OD, 11 days	10 mg, single dose	2.3-fold ↑
Glecaprevir 400 mg/pibrentasvir 120 mg OD, 7 days	5 mg OD, 7 days	2.2-fold ↑
Lopinavir 400 mg/ritonavir 100 mg BID, 17 days	20 mg OD, 7 days	2.1-fold ↑
Capmatinib 400mg BID	10 mg, single dose	2.1-fold ↑
Clopidogrel 300 mg loading, followed by 75 mg at 24 hours	20 mg, single dose	2-fold ↑
Fostamatinib 100 mg twice daily	20 mg, single dose	2.0-fold ↑
Tafamidis 61 mg BID on Days 1 & 2, Followed by OD on Days 3 to 9	10 mg, single dose	2.0-fold ↑
Less than 2-fold increase in AUC of rosuvastatin		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Febuxostat 120 mg OD	10 mg, single dose	1.9-fold ↑
Gemfibrozil 600 mg BID, 7 days	80 mg, single dose	1.9-fold ↑
Eltrombopag 75 mg OD, 5 days	10 mg, single dose	1.6-fold ↑
Darunavir 600 mg/ritonavir 100 mg BID, 7 days	10 mg OD, 7 days	1.5-fold ↑
Tipranavir 500 mg/ritonavir 200 mg BID, 11 days	10 mg, single dose	1.4-fold ↑
Dronedarone 400 mg BID	Not available	1.4-fold ↑
Itraconazole 200 mg OD, 5 days	10 mg, single dose	1.4-fold ↑**
Ezetimibe 10 mg OD, 14 days	10 mg, OD, 14 days	1.2-fold ↑**
Decrease in AUC of rosuvastatin		

Table 1. Effect of co-administered medicinal products on rosuvastatin exposure (AUC; in order of decreasing magnitude) from published clinical trials		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC*
Erythromycin 500 mg QID, 7 days	80 mg, single dose	20 % ↓
Baicalin 50 mg TID, 14 days	20 mg, single dose	47 % ↓
* Data given as x-fold change represent a simple ratio between co-administration and rosuvastatin alone. Data given as % change represent % difference relative to rosuvastatin alone. Increase is indicated as “↑”, no change as “↔”, decrease as “↓”. ** Several interaction studies have been performed at different rosuvastatin dosages, the table shows the most significant ratio OD = once daily; BID = twice daily; TID = three times daily; QID = four times daily		

The following medical product/combinations did not have a clinically significant effect on the AUC ratio of rosuvastatin at coadministration: Aleglitazar 0.3 mg 7 days dosing; Fenofibrate 67 mg 7 days TID dosing; Fluconazole 200mg 11 days OD dosing; Fosamprenavir 700 mg/ritonavir 100 mg 8 days BID dosing; Ketoconazole 200 mg 7 days BID dosing; Rifampin 450 mg 7 days OD dosing; Silymarin 140 mg 5 days TID dosing.

Effect of rosuvastatin on co-administered medicinal products

Vitamin K antagonists: As with other HMG-CoA reductase inhibitors, the initiation of treatment or dosage up-titration of rosuvastatin in patients treated concomitantly with vitamin K antagonists (e.g. warfarin or another coumarin anticoagulant) may result in an increase in International Normalised Ratio (INR). Discontinuation or down-titration of rosuvastatin may result in a decrease in INR. In such situations, appropriate monitoring of INR is desirable.

Oral contraceptive/hormone replacement therapy (HRT): Concomitant use of rosuvastatin and an oral contraceptive resulted in an increase in ethinyl estradiol and norgestrel AUC of 26% and 34%, respectively. These increased plasma levels should be considered when selecting oral contraceptive doses. There are no pharmacokinetic data available in subjects taking concomitant rosuvastatin and HRT and therefore a similar effect cannot be excluded. However, the combination has been extensively used in women in clinical trials and was well tolerated.

Other medicinal products

Digoxin: Based on data from specific interaction studies no clinically relevant interaction with digoxin is expected.

Fusidic Acid: Interaction studies with rosuvastatin and fusidic acid have not been conducted. The risk of myopathy including rhabdomyolysis may be increased by the concomitant administration of systemic fusidic acid with statins. The mechanism of this interaction (whether it is pharmacodynamic or pharmacokinetic, or both) is yet unknown. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving this combination.

If treatment with systemic fusidic acid is necessary, rosuvastatin treatment should be discontinued throughout the duration of the fusidic acid treatment. Also see section 4.4

Linked to acetylsalicylic acid component

The use of several platelet aggregation inhibitors, i.e. ASA, NSAIDs, ticlopidine, clopidogrel, tirofiban, eptifibatide, increases the risk of bleeding, likewise their combination with heparin and its derivatives (hirudine, fondaparinux), oral anticoagulants and thrombolytics. Clinical and biological parameters of haemostasis should be regularly monitored in patients about to receive thrombolytic therapy.

Contraindicated combinations

Methotrexate (used at doses > 15 mg/week): the combined drugs, methotrexate and ASA, increase haematological toxicity of methotrexate due to the decreased renal clearance of methotrexate by ASA. Therefore, the concomitant use of methotrexate with ZAHRON ASA is contraindicated (see section 4.3).

Not recommended associations

- *Uricosurics agents* (benzbromarone, probenecid, and sulphinpyrazone): reduced effect of uric acid excretion by competition of renal tubular uric acid elimination. Therefore, the concomitant use of ZAHRON ASA with uricosurics agents is not recommended (see section 4.4).
- *Valproic acid*: increased risk of adverse effects due displacement from its plasma protein binding by salicylates.
- *Selective serotonin reuptake inhibitors (SSRIs)*: increased risk of gastrointestinal bleeding due to synergistic effects.
- *Digoxin*: increase in plasma concentration
- *Antidiabetics* such as insulin, sulfonylureas in combination with acetylsalicylic acid in higher doses: increased risk of hypoglycaemic effect.

Combinations requiring precautions for use

- *Diuretics*: risk of acute renal failure due to the decreased glomerular filtration via decreased renal prostaglandin synthesis. Hydrating the patient and monitoring renal function at the start of the treatment. Decrease in effect may be caused by aldosterone antagonists (spironolactone and potassium canrenoate) or loop diuretics (e.g. furosemide).
- *Systemic glucocorticosteroids*: the concomitant administration of steroids (except hydrocortisone used as replacement therapy in Addison's disease): increased risk of gastrointestinal adverse effects.
- *Methotrexate used at doses lower than 15 mg/week*: the combined drugs, methotrexate and ASA, increased haematological toxicity of methotrexate due to the decreased renal clearance of methotrexate by ASA. Weekly blood count checks should be done during the first weeks of the combination. Enhanced monitoring in the presence of even mildly impaired renal function, as well as in elderly.
- *Metamizole* may reduce the effect of acetylsalicylic acid on platelet aggregation, when taken concomitantly. Therefore, this combination should be used with caution in patients taking low dose acetylsalicylic acid for cardioprotection.
- *Heparin used at curative dosage or in elderly patients*: when ASA is coadministered with heparin at curative dosage or in elderly patients, there is an increased risk of bleeding. Close monitoring of the INR, aPTT and/or bleeding time should be performed in the case of concomitant administration of both drugs.
- *NSAIDs*: increased risk of bleeding and of damage on gastrointestinal mucosa and enhancement of prolonging bleeding time.

Combinations to be taken into account

- *Other anticoagulants* (coumarin derivatives, heparin at preventive dosage), acetylsalicylic acid can increase the risk of bleeding if administered prior to thrombolytic therapy. Patients about to receive thrombolytic therapy must therefore be closely monitored for signs of external or internal bleeding (e.g. bruises).

- *Other platelet antiaggregants and other thrombolytics* (e.g. ticlopidine, clopidogrel): increased risk of bleeding due to prolongation of clotting time.
- NSAIDs: Co-administration (same day) of some NSAIDs (excluding acetylsalicylic acid), such as ibuprofen and naproxen, may attenuate the irreversible anti-platelet effect of acetylsalicylic acid. The clinical relevance of this interaction is unknown. Treatment of patients with an increased cardiovascular risk with some NSAIDs, such as ibuprofen or naproxen, may limit the cardioprotective effect of acetylsalicylic acid.
- Antihypertensives (in particular ACE inhibitors)
- *Antacids*: antacids can increase the renal excretion of ASA by alkalinising the urine.
- *Alcohol*: addition of their own damage on gastrointestinal mucosa and enhancement of prolonging bleeding time.

Experimental data suggest that ibuprofen may inhibit the effect of low dose acetylsalicylic acid on platelet aggregation when administered concomitantly with acetylsalicylic acid. However, the limitations of these data and the uncertainties regarding extrapolation of *ex vivo* data to the clinical situation imply that no firm conclusions can be made for regular ibuprofen use. No clinically relevant effect is considered to be likely for occasional ibuprofen use (see section 5.1)

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

Women of childbearing potential must use appropriate contraceptive measures during treatment with ZAHRON ASA (see section 4.3).

Pregnancy

ZAHRON ASA is contraindicated in pregnancy (see section 4.3).

If a patient becomes pregnant during use of this product, treatment should be discontinued immediately.

Rosuvastatin

Since cholesterol and other products of cholesterol biosynthesis are essential for the development of the foetus, the potential risk from inhibition of HMG-CoA reductase outweighs the advantage of treatment during pregnancy. Animal studies provide limited evidence of reproductive toxicity of rosuvastatin (see section 5.3).

Acetylsalicylic acid

The safety of acetylsalicylic acid in human pregnancy has not been established. Inhibition of prostaglandin synthesis may adversely affect the pregnancy and/or embryo/foetal development. Data from epidemiological studies suggest there is an increased risk of miscarriage and cardiac malformations and gastroschisis following the use of a prostaglandin synthesis inhibitor in early pregnancy.

Low doses (up to and including 100 mg/day):

Clinical studies indicate that doses up to 100 mg/day for restricted obstetrical use, which require specialised monitoring, appear safe.

Doses of above 100 mg/day and up to 500 mg/day:

There is insufficient clinical experience regarding the use of doses above 100 mg/day up to 500

mg/day. Therefore, the recommendations below for doses of 500 mg/day and above apply also for this dose range.

Doses of 500 mg/day and above

From the 20th week of pregnancy onward, <x> use may cause oligohydramnios resulting from foetal renal dysfunction. This may occur shortly after treatment initiation and is usually reversible upon discontinuation. In addition, there have been reports of ductus arteriosus constriction following treatment in the second trimester, most of which resolved after treatment cessation.

During the third trimester of pregnancy, all prostaglandin synthesis inhibitors may expose the foetus to:

- cardiopulmonary toxicity (premature constriction/closure of the ductus arteriosus and pulmonary hypertension);
- renal dysfunction (see above);

the mother and the neonate, at the end of pregnancy, to:

- possible prolongation of bleeding time, an anti-aggregating effect which may occur even at very low doses;
- inhibition of uterine contractions resulting in delayed or prolonged labour.

Fertility

Rosuvastatin

There are no known effects on fertility after use of rosuvastatin.

Acetylsalicylic acid

In animal studies reproductive toxicity was observed at high doses of acetylsalicylic acid (see section 5.3).

Breast-feeding

ZAHRON ASA is contraindicated during breast-feeding (see section 4.3).

Rosuvastatin

Limited data from published reports indicate that rosuvastatin is present in human milk. Rosuvastatin is excreted in the milk of rats. Due to rosuvastatin's mechanism of action, there is a potential risk for adverse reactions in the infant.

Acetylsalicylic acid

Small amounts of acetylsalicylic acid and its degradation products also enter breast milk.

4.7 Effects on ability to drive and use machines

ZAHRON ASA can have minor or moderate influence on the ability to drive and use machines.

Studies to determine the effect of rosuvastatin on the ability to drive and use machines have not been conducted. However, based on its pharmacodynamic properties, rosuvastatin is unlikely to affect this ability. When driving vehicles or operating machines, it should be taken into account that dizziness may occur during treatment.

Acetylsalicylic acid has no influence on the ability to drive or use machines.

4.8 Undesirable effects

Side effects are grouped on the basis of System Organ Class. Within each system organ class the

frequencies are defined as:

- Very common ($\geq 1/10$)
- Common ($\geq 1/100$ to $< 1/10$)
- Uncommon ($\geq 1/1,000$ to $< 1/100$)
- Rare ($\geq 1/10,000$ to $< 1/1,000$)
- Very rare ($< 1/10,000$)
- Not known (cannot be estimated from the available data)

Tabulated list of adverse reactions

MedDRA System Organ Class	Undesirable Effects	Frequency	
		Rosuvastatin	Acetylsalicylic acid
Blood and lymphatic system disorders	Bleeding such as nose bleed, gingival bleeding, ecchymosis or bleeding in the urogenital tract with a possible prolongation of clotting time (see section 4.4)		Common
	Intracranial bleeding, blood in urine. Iron deficiency anaemia due to occult blood losses from the gastrointestinal tract may occur after long-term use of acetylsalicylic acid 100 mg		Uncommon
	Severe bleeding such as cerebral haemorrhaging, particularly in patients with uncompensated high blood pressure and/or concomitant treatment with anticoagulants, which may be life-threatening in individual cases		Rare
	Thrombocytopenia	Rare	-
	Haemolysis and haemolytic anaemia in patients with severe glucose-6-phosphate dehydrogenase deficiency.		Not known
Immune system disorders	Hypersensitivity reactions of the skin, respiratory tract, the gastrointestinal tract and the cardiovascular system, especially in patients with asthma. Possible symptoms are: lowered blood pressure, attacks of breathlessness, rhinitis, nasal congestion, anaphylactic shock or Quincke's oedema.	-	Rare
	Hypersensitivity reactions including angioedema	Rare	-
Metabolism and nutrition disorders	Hypoglycaemia	-	Very rare
	Reduced excretion of uric acid leading to attacks of gout in susceptible patients		Very rare
Endocrine disorders	Diabetes mellitus ¹	Common	-
Psychiatric disorders	Depression	Not known	-

	Mental confusion	-	Rare
Nervous system disorders	Dizziness, headache	Common	Rare
	Polyneuropathy, memory loss	Very rare	-
	Peripheral neuropathy. Sleep disorders (insomnia, nightmares),	Not known	
	Myasthenia gravis	Not known	-
Eye disorders	Ocular myasthenia	Not known	-
Ear and labyrinth disorders	Impaired hearing or tinnitus	-	Rare
Respiratory, thoracic and mediastinal disorders	Cough, dyspnoea	Not known	-
Gastrointestinal disorders	Heartburn	-	Common
	Vomiting		Common
	Abdominal pain	Common	Common
	Nausea	Common	Common
	Minor blood loss from gastrointestinal tract (micro-bleeding).		Common
	Constipation	Common	-
	Pancreatitis	Rare	-
	Diarrhoea	Not known	Common
	Gastrointestinal ulcers, which may in very rare cases lead to perforation		Uncommon
	Gastrointestinal bleeding		Uncommon
	Gastrointestinal inflammation		Uncommon
	In the case of previous damage to the intestinal mucosa, multiple membranes can form in the intestinal lumen with potential subsequent stenosis		Not known
Hepatobiliary disorders	Hepatitis	Very rare	-
	Jaundice	Very rare	-
	Increased hepatic transaminases	Rare	-
	Elevated liver function values		Very rare
Skin and subcutaneous tissue disorders	Skin reactions		Uncommon
	Rash	Uncommon	-
	Pruritus	Uncommon	-
	Urticaria	Uncommon	-
	Stevens-Johnson syndrome	Not known	-
	Erythema multiforme		Very rare
	Drug reaction with eosinophilia and systemic symptoms (DRESS)	Not known	
Musculoskeletal and	Myalgia	Common	-

connective tissue disorders	Myopathy (including myositis), Rhabdomyolysis, lupus like syndrome, muscle rupture	Rare	-
	Arthralgia	Very rare	-
	Immune-mediated necrotising myopathy	Not known	-
	Tendon disorders, sometimes complicated by rupture	Not known	-
Renal and urinary disorders	Impaired renal function and acute renal failure	-	Very rare
	Haematuria	Very rare	
Reproductive system and breast disorders	Gynaecomastia	Very rare	-
General disorders and administration site conditions	Asthenia	Common	-
	Oedema	Not known	-

¹ Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI >30 kg/m², raised triglycerides, history of hypertension).

As with other HMG-CoA reductase inhibitors, the incidence of adverse drug reactions tends to be dose dependent.

Description of selected adverse reactions associated with rosuvastatin

Renal Effects: Proteinuria, detected by dipstick testing and mostly tubular in origin, has been observed in patients treated with rosuvastatin. Shifts in urine protein from none or trace to ++ or more were seen in $<1\%$ of patients at some time during treatment with 10 and 20 mg, and in approximately 3% of patients treated with 40 mg. A minor increase in shift from none or trace to/was observed with the 20 mg dose. In most cases, proteinuria decreases or disappears spontaneously on continued therapy. Review of data from clinical trials and post-marketing experience to date has not identified a causal association between proteinuria and acute or progressive renal disease.

Haematuria has been observed in patients treated with rosuvastatin and clinical trial data show that the occurrence is low.

Skeletal muscle effects: Effects on skeletal muscle e.g. myalgia, myopathy (including myositis) and, rarely, rhabdomyolysis with and without acute renal failure have been reported in rosuvastatin-treated patients with all doses and in particular with doses >20 mg.

A dose-related increase in CK levels has been observed in patients taking rosuvastatin; the majority of cases were mild, asymptomatic and transient. If CK levels are elevated ($>5\times$ ULN), treatment should be discontinued (see section 4.4).

Liver Effects: As with other HMG-CoA reductase inhibitors, a dose-related increase in transaminases has been observed in a small number of patients taking rosuvastatin; the majority of cases were mild, asymptomatic and transient.

The following adverse events have been reported with some statins:

- Sexual dysfunction
- Exceptional cases of interstitial lung disease, especially with long term therapy (see section 4.4)

The reporting rates for rhabdomyolysis, serious renal events and serious hepatic events (consisting mainly of increased hepatic transaminases) is higher at the 40 mg dose.

Paediatric population: Creatine kinase elevations >10xULN and muscle symptoms following exercise or increased physical activity were observed more frequently in a 52-week clinical trial of children and adolescents compared to adults (see section 4.4). In other respects, the safety profile of rosuvastatin was similar in children and adolescents compared to adults.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions directly to the National Centre for Adverse Drug Reaction Monitoring by visiting the website npra.gov.my [Consumers → Reporting Side Effects to Medicines (ConSERF) or Vaccines (AEFI)].

4.9 Overdose

Symptoms

Rosuvastatin

To date there has been limited experience with overdosage of rosuvastatin.

There is no specific treatment in the event of overdose. The patient should be treated symptomatically and supportive measures instituted as required. Liver function and CK levels should be monitored. Haemodialysis is unlikely to be of benefit.

Acetylsalicylic acid

Overdosage is unlikely due to the low level of ASA in ZAHRON ASA. However intoxication (accidental overdose) in very young children or therapeutic overdose in elderly may present as follows: The following are associated with moderate intoxication: dizziness, headache, tinnitus, confusion and gastrointestinal symptoms (nausea, vomiting and gastric pain).

With severe intoxication, serious disturbances of the acid-base equilibrium occur. Initial hyperventilation leads to respiratory alkalosis. Subsequently a respiratory acidosis occurs as a result of a suppressive effect on the respiratory centre. A metabolic acidosis also arises due to the presence of salicylate. Given that children, infants and toddlers are often only seen at a late stage of intoxication, they will usually have already reached the acidosis stage.

The following can also arise: hyperthermia and perspiration, leading to dehydration, restlessness, convulsions, hallucinations and hypoglycaemia. Depression of the nervous system can lead to coma, cardiovascular collapse and respiratory arrest. The lethal dose of ASA is 25-30 gram. Plasma salicylate concentrations above 300 mg/l (1,67 mmol/l) suggest intoxication.

Management

If overdose occurs, the treatment with ZAHRON ASA should be stopped and supportive and symptomatic treatment should be provided.

If a toxic dose has been ingested then admission to hospital is necessary. With moderate intoxication an attempt can be made to induce vomiting; if this fails, gastric lavage is indicated. Activated charcoal (adsorbent) and sodium sulphate (laxative) are then administered. Alkalinising of the urine (250 mmol NaHCO₃ for 3 hours) while monitoring the urine pH is indicated. Haemodialysis is the preferred treatment for severe intoxication.

Liver function and CK levels should be monitored.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: lipid modifying agents, combinations, ATC code: C10BX05.

Rosuvastatin

Rosuvastatin is a selective and competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3- hydroxy-3-methylglutaryl coenzyme A to mevalonate, a precursor for cholesterol. The primary site of action of rosuvastatin is the liver, the target organ for cholesterol lowering.

Rosuvastatin increases the number of hepatic LDL receptors on the cell-surface, enhancing uptake and catabolism of LDL and it inhibits the hepatic synthesis of VLDL, thereby reducing the total number of VLDL and LDL particles.

Rosuvastatin reduces elevated LDL-cholesterol, total cholesterol and triglycerides and increases HDL-cholesterol. It also lowers ApoB, nonHDL-C, VLDL-C, VLDL-TG and increases ApoA-I . Rosuvastatin also lowers the LDL-C/HDL-C, total C/HDL-C and nonHDL-C/HDL-C and the ApoB/ApoA-I ratios.

Table 2: Dose response in patients with primary hypercholesterolaemia (type IIa and IIb)
(adjusted mean percent change from baseline)

Dose	N	LDL-C	Total-C	HDL-C	TG	nonHDL-C	ApoB	ApoA-I
Placebo	13	-7	-5	3	-3	-7	-3	0
5 mg	17	-45	-33	13	-35	-44	-38	4
10 mg	17	-52	-36	14	-10	-48	-42	4
20 mg	17	-55	-40	8	-23	-51	-46	5
40 mg	18	-63	-46	10	-28	-60	-54	0

A therapeutic effect is obtained within 1 week following treatment initiation and 90% of maximum response is achieved in 2 weeks. The maximum response is usually achieved by 4 weeks and is maintained after that.

Acetylsalicylic acid

Acetylsalicylic acid irreversibly inhibits platelet aggregation. This effect on platelets is due to cyclooxygenase acetylation. This irreversibly inhibits synthesis of thromboxane A₂ (a platelet aggregation promoting and vasoconstricting prostaglandin) in the platelets. This effect is permanent and usually lasts for the entire 8-day life span of a platelet.

Paradoxically, acetylsalicylic acid also inhibits the synthesis of prostacyclin (a platelet aggregation inhibiting prostaglandin but with vasodilatory effects) in the endothelial cells of the blood vessels. This effect is temporary.

As soon as the acetylsalicylic acid has been eliminated from the blood, the nucleated endothelial cells synthesize prostacyclin again.

As a result, a single, low daily dose of acetylsalicylic acid (< 100 mg/day) results in inhibition of thromboxane A₂ in the platelets without considerably affecting prostacyclin synthesis.

5.2 Pharmacokinetic properties

Absorption, distribution

Rosuvastatin

Maximum rosuvastatin plasma concentrations are achieved approximately 5 hours after oral administration. The absolute bioavailability is approximately 20%. Rosuvastatin is taken up extensively by the liver which is the primary site of cholesterol synthesis and LDL-C clearance. The volume of distribution of rosuvastatin is approximately 134 l. Approximately 90% of rosuvastatin is bound to plasma proteins, mainly to albumin.

Genetic polymorphisms: Disposition of HMG-CoA reductase inhibitors, including rosuvastatin, involves OATP1B1 and BCRP transporter proteins. In patients with SLCO1B1 (OATP1B1) and/or ABCG2 (BCRP) genetic polymorphisms there is a risk of increased rosuvastatin exposure. Individual polymorphisms of SLCO1B1 c.521CC and ABCG2 c.421AA are associated with a higher rosuvastatin exposure (AUC) compared to the SLCO1B1 c.521TT or ABCG2 c.421CC genotypes. This specific genotyping is not established in clinical practice, but for patients who are known to have these types of polymorphisms, a lower daily dose of rosuvastatin is recommended.

Acetylsalicylic acid

After oral administration, ASA is rapidly absorbed in the proximal part of the small bowel. The maximum plasma concentration is reached after 0.5-2 hours. However, a considerable part of the dose is hydrolysed in the gastric wall during absorption.

Absorption of ASA is generally rapid and complete following oral administration. Food decreases the rate, but not the extent, of absorption of acetylsalicylic acid.

The volume of distribution of ASA is approximately 0.20 l/kg bodyweight. The first conversion product formed from ASA, anti-inflammatorily effective salicylic acid, is bound to plasma proteins, primarily albumin, to the 90% level.

Salicylic acid diffuses slowly to the synovia and the synovial fluid. It penetrates the placenta and passes over into the maternal milk.

Biotransformation, elimination

Rosuvastatin

Rosuvastatin undergoes limited metabolism (approximately 10%). *In vitro* metabolism studies using human hepatocytes indicate that rosuvastatin is a poor substrate for cytochrome P450-based metabolism. CYP2C9 was the principal isoenzyme involved, with 2C19, 3A4 and 2D6 involved to a lesser extent. The main metabolites identified are the N-desmethyl and lactone metabolites. The N-desmethyl metabolite is approximately 50% less active than rosuvastatin whereas the lactone form is considered clinically inactive. Rosuvastatin accounts for greater than 90% of the circulating HMG-CoA reductase inhibitor activity. Approximately 90% of the rosuvastatin dose is excreted unchanged in the faeces (consisting of absorbed and non-absorbed active substance) and the remaining part is excreted in urine. Approximately 5% is excreted unchanged in urine. The plasma elimination half-life is approximately 20 hours. The elimination half-life does not increase at higher doses. The geometric mean plasma clearance is approximately 50 litres/hour (coefficient of variation 21.7%). As with other HMG-CoA reductase inhibitors, the hepatic uptake of rosuvastatin involves the membrane transporter OATP-C. This transporter is important in the hepatic elimination of rosuvastatin.

Acetylsalicylic acid

ASA is primarily converted into salicylic acid by hydrolysis. The half-life of ASA is short, approx. 15-20 minutes.

Salicylic acid is then converted into glycine acid and glucuronic acid conjugates and traces of gentisinic acid. At higher therapeutic doses the conversion capacity of salicylic acid is already exceeded and the pharmacokinetics is non-linear. This leads to a prolongation of the apparent elimination half-life of salicylic acid from a few hours to approximately 24 hours.

Excretion is primarily via the kidneys.

The tubular resorption of salicylic acid is pH-dependent. By alkalising the urine the proportion of unchanged salicylic acid in the excretion increases from approx. 10% to approximately 80%.

Linearity/non-linearity

Systemic exposure of rosuvastatin increases in proportion to dose. There are no changes in pharmacokinetic parameters following multiple daily doses.

Special populations:

Age and sex

There was no clinically relevant effect of age or sex on the pharmacokinetics of rosuvastatin in adults. The pharmacokinetics of rosuvastatin in children and adolescents with heterozygous familial hypercholesterolaemia was similar to that of adult volunteers.

Race

Pharmacokinetic studies show an approximate 2-fold elevation in median AUC and $C_{\max}$ of rosuvastatin in Asian subjects (Japanese, Chinese, Filipino, Vietnamese and Koreans) compared with Caucasians; Asian-Indians show an approximate 1.3-fold elevation in median AUC and $C_{\max}$. A population pharmacokinetic analysis revealed no clinically relevant differences in pharmacokinetics between Caucasian and Black groups.

Renal impairment

In a study in subjects with varying degrees of renal impairment, mild to moderate renal disease had no influence on plasma concentration of rosuvastatin or the N-desmethyl metabolite. Subjects with severe impairment ($CK < 30$ ml/min) had a 3-fold increase in plasma concentration and a 9-fold increase in the N-desmethyl metabolite concentration compared to healthy volunteers. Steady-state plasma concentrations of rosuvastatin in subjects undergoing haemodialysis were approximately 50% greater compared to healthy volunteers.

Hepatic impairment

In a study with subjects with varying degrees of hepatic impairment there was no evidence of increased exposure to rosuvastatin in subjects with Child-Pugh scores of 7 or below. However, two subjects with Child-Pugh scores of 8 and 9 showed an increase in systemic exposure of at least 2-fold compared to subjects with lower Child-Pugh scores. There is no experience in subjects with Child-Pugh scores above 9.

5.3 Preclinical safety data

Rosuvastatin

Preclinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, genotoxicity and carcinogenicity potential. Specific tests for effects on hERG have not been evaluated. Adverse reactions not observed in clinical studies, but seen in animals at exposure levels similar to clinical exposure levels were as follows: In repeated-dose toxicity studies histopathologic liver changes likely due to the pharmacologic action of rosuvastatin were observed in mouse, rat, and to a lesser extent with effects in the gall bladder in dogs, but not in monkeys. In addition, testicular toxicity was observed in monkeys and dogs at higher dosages. Reproductive toxicity was evident in rats, with reduced litter sizes, litter weight and pup survival observed at maternally toxic doses, where systemic exposures were several times above the therapeutic exposure level.

Acetylsalicylic acid

The preclinical safety profile of acetylsalicylic acid is well documented. In animal studies, salicylates have shown no other organ injury than renal damage. Acetylsalicylic acid has been extensively examined for possible mutagenic and carcinogenic effects; no relevant evidence of mutagenic or carcinogenic potential effects has been determined.

It has been reported that salicylates have teratogenic effects on several animal species (such as cardiac and skeletal malformation, gastroschisis). Impaired implantation, embryo- and foetotoxic effects and impaired learning ability have been described in offspring exposed during gestation.

6 PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Film-coated tablet with rosuvastatin:

Tablet core

Lactose monohydrate
Cellulose, microcrystalline
Magnesium oxide, heavy
Crospovidone (type A)
Silica, colloidal anhydrous
Magnesium stearate

Film coating system

Polyvinyl alcohol
Titanium dioxide (E 171)
Talc
Iron oxide yellow (E 172)
Lecithin (soya)
Iron oxide red (E 172)
Xanthan gum
Black iron oxide (E 172)

Tablet with acetylsalicylic acid:

Cellulose, microcrystalline
Maize starch
Silica, colloidal anhydrous
Stearic acid

Capsule shell:

Gelatin
Titanium dioxide (E 171)
Indigotine (E 132)
Yellow iron oxide (E 172)

Black ink:

Shellac
Propylene glycol
Strong ammonia solution
Black iron oxide (E 172)
Potassium hydroxide

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

3 years

6.4 Special precautions for storage

Store below 30°C.

Store in the original package in order to protect from moisture.

6.5 Nature and contents of container

Blister Polyamide-Aluminium-PVC (laminate) and aluminium lidding foil: 10, 28, 30, 56, 60, 90, 100

capsules, in a box.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7 PRODUCT REGISTRATION HOLDER

DKSH Malaysia Sdn Bhd
B-11-01 The Ascent Paradigm,
No. 1 Jalan SS7/26A Kelana Jaya,
47301 Petaling Jaya, Selangor,
Malaysia

8 PRODUCT REGISTRATION NUMBER

MAL25126003AZ - ZAHRON ASA (Rosuvastatin 5mg/Acetylsalicylic acid 100mg) Hard Capsule
MAL25126004AZ - ZAHRON ASA (Rosuvastatin 10mg/Acetylsalicylic acid 100mg) Hard Capsule
MAL25126005AZ - ZAHRON ASA (Rosuvastatin 20mg/Acetylsalicylic acid 100mg) Hard Capsule

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